CS 59300
Application of Deep Learning
ChatGPT APIs

Instructor: Dr. Zesheng Chen

Acknowledgements

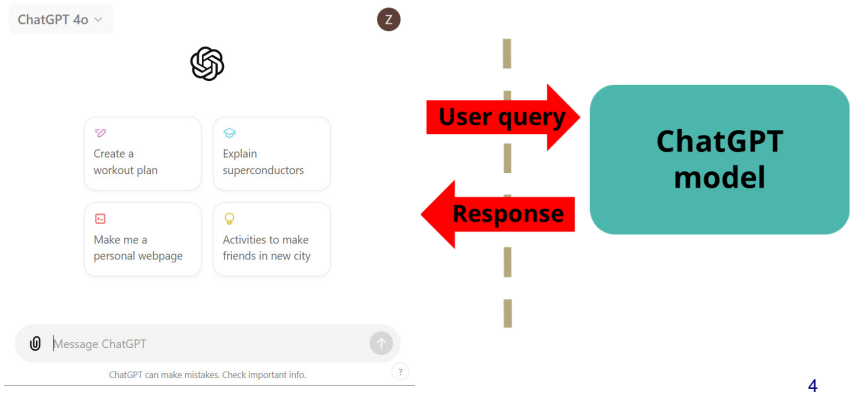
- Refer to the following:
  - Introduction to Generative AI 2024 Spring
  - https://speech.ee.ntu.edu.tw/~hylee/genai/2024-spring.php
  - T81-559: Applications of Generative Artificial Intelligence
  - https://github.com/jeffheaton/app_generative_ai

Outline

- Motivation
- Setup
- Examples

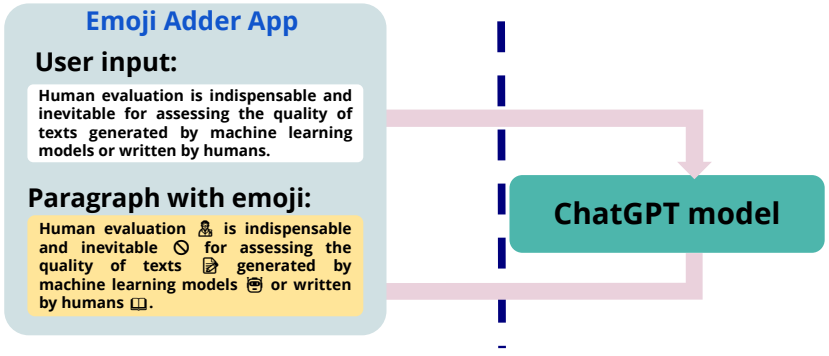
Motivation

- Using ChatGPT model by browser



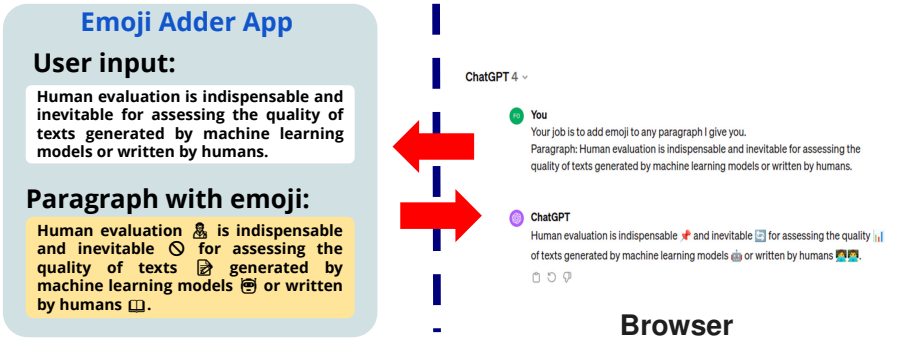
Creating Your Own Application Using ChatGPT

For example, we want to create an application that adds appropriate emojis to any given input, and you want to power it by ChatGPT.



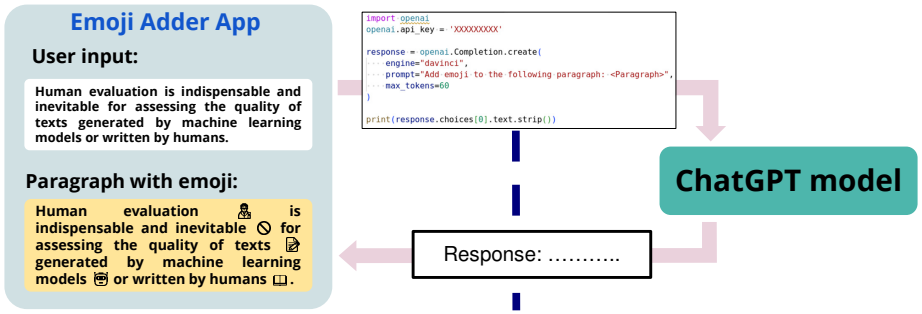
Creating Your Own Application Using ChatGPT

While you can manually copy the user input to the browser and copy the response from ChatGPT to the user end, this is too slow



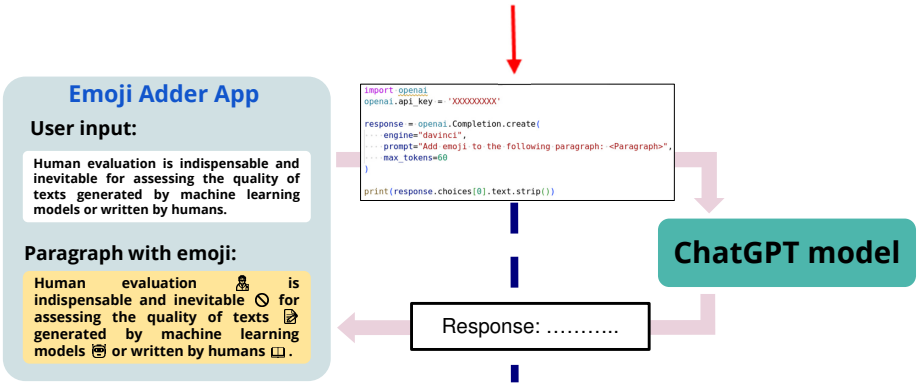
Creating Your Own Application Using ChatGPT API

The ChatGPT API (Application Programming Interface) enables programmatic interaction with the ChatGPT model



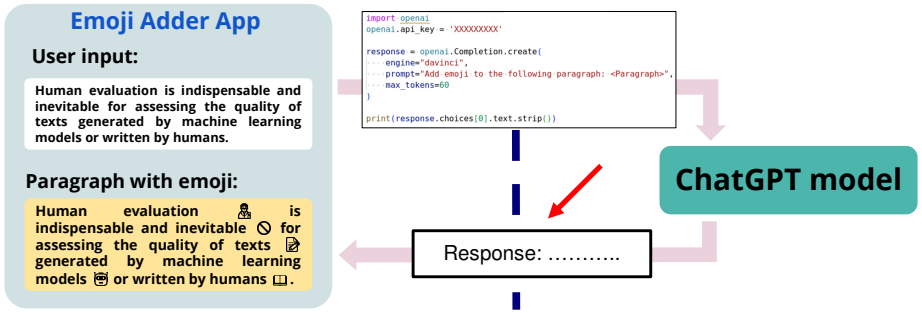
Creating Your Own Application Using ChatGPT API

You tell ChatGPT what you want it to do in a programmatic way



Creating Your Own Application Using ChatGPT API

ChatGPT returns the response by following you what you ask it to do



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Setup

- https://openai.com/api/
- Sign up for an OpenAI account for API.
- Must add a payment method and some credits to your account.
  - Cannot use ChatGPT APIs without adding the credit card.
- Can start with a small amount of credits, maybe $10, and add more as needed.
- Create an "API key".
  - Save this secret key somewhere safe and accessible.

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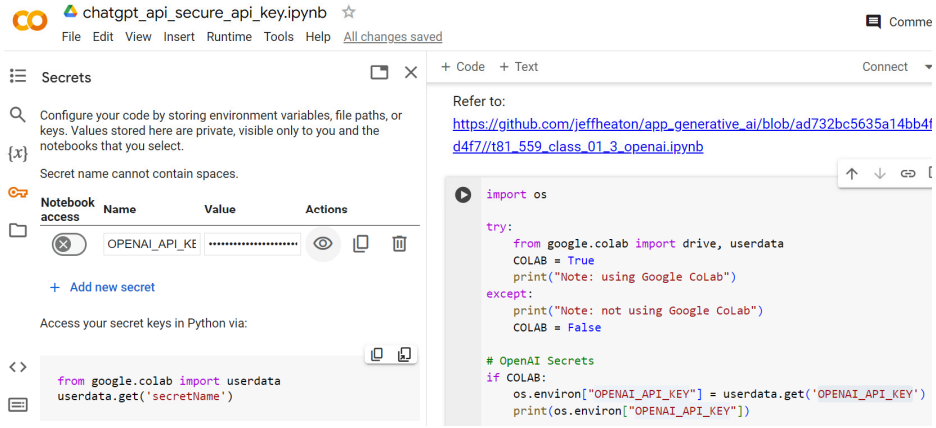
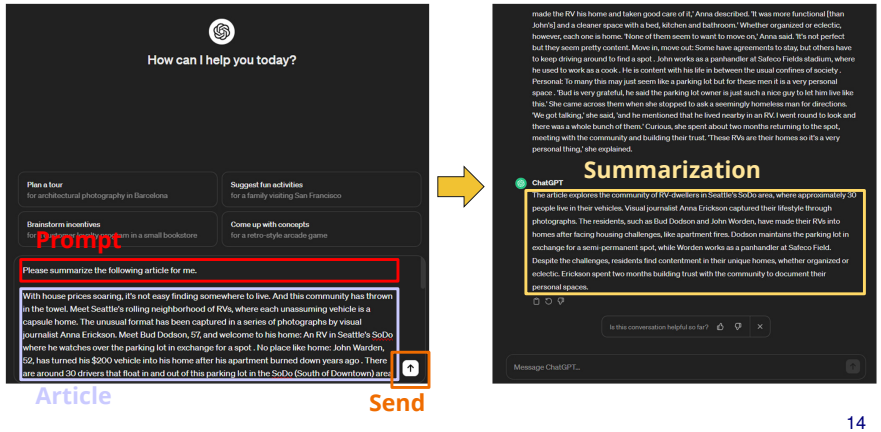
Securing API Key at Google CoLab

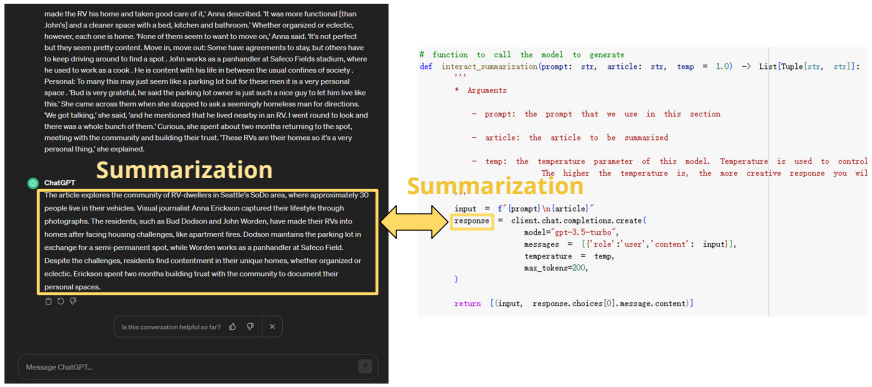


Use Web-based ChatGPT

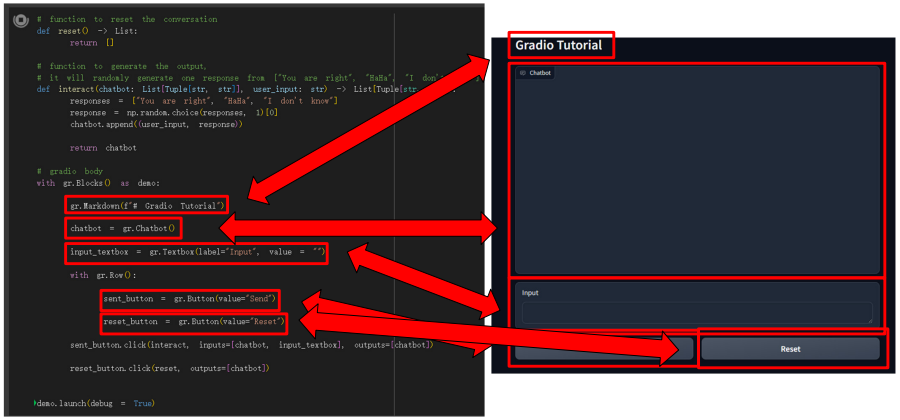


Web-based vs. API-based

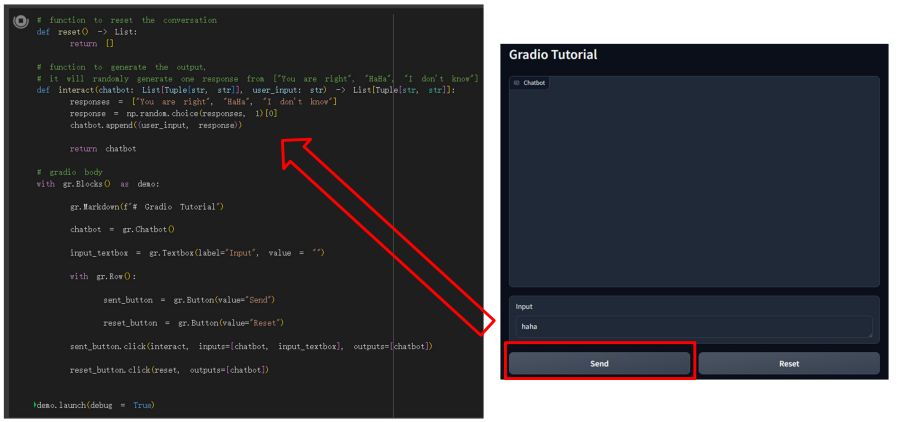
Web-based vs. API-based



Gradio UI

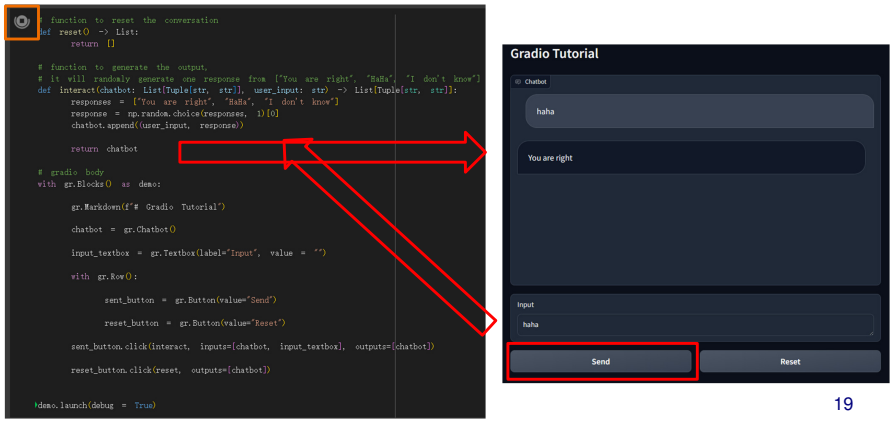


Gradio UI



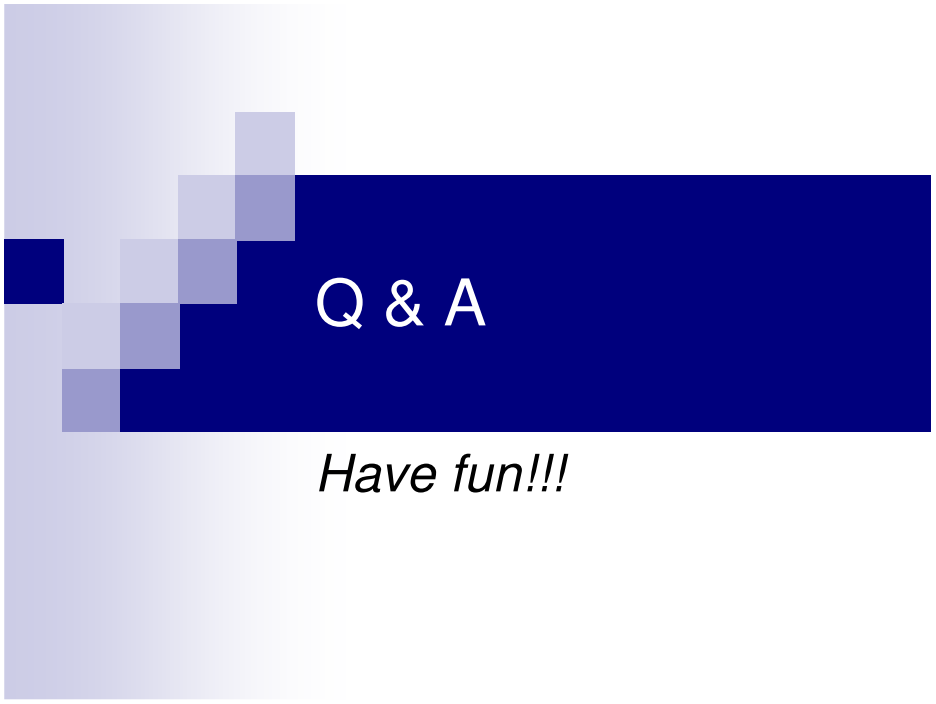
Gradio UI

When it's spinning, the gradio is connected to the colab.
When it stops, the gradio will no longer work.



Summary

- Learn how to obtain ChatGPT API key
- Learn how to call the ChatGPT API
- Learn how to design the prompts to achieve the functionality you want



Have fun!!!